

FORM TP 2013177



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MAY/JUNE 2013

CARIBBEAN EXAMINATIONS COUNCIL
CARIBBEAN ADVANCED PROFICIENCY EXAMINATION®
ENVIRONMENTAL SCIENCE
ECOLOGY, HUMAN POPULATION AND NATURAL RESOURCES

UNIT 1 – Paper 02

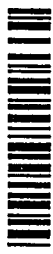
2 hours 30 minutes

07 MAY 2013 (p.m.)

READ THE FOLLOWING INSTRUCTIONS CAREFULLY.

1. Do not open this examination paper until instructed to do so.
2. This paper consists of **SIX** questions, **TWO** from each Module.
3. Candidates **MUST** answer **ALL** questions.
4. Write your answers in the answer booklet provided.
5. The use of a silent, non-programmable, scientific calculator is permitted.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO.



MODULE 1

Answer BOTH questions.

1. (a) Figure 1 is a diagrammatic representation of the nitrogen cycle.

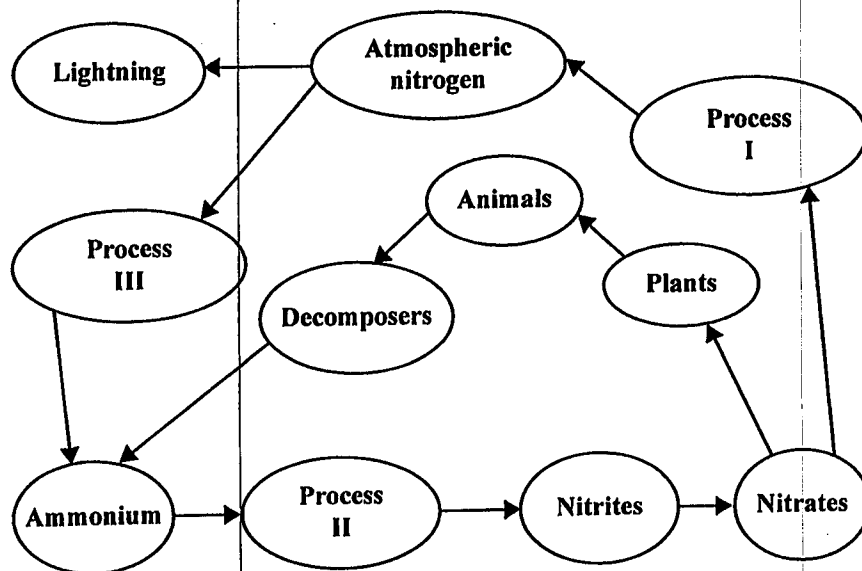


Figure 1. The nitrogen cycle

- (i) Name the types of bacteria that allow processes I, II and III to occur.

Process	Type of Bacteria
I	
II	
III	

[3 marks]

- (ii) Explain THREE ways by which human beings may alter the nitrogen cycle.

[6 marks]

- (iii) A farmer was advised that he could reduce the amount of money spent on nitrogen fertilizers if he practised intercropping with legumes regularly. Justify the advice given to the farmer.

[5 marks]

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- (b) Table 1 below shows the results of an ecological study to determine the diversity of insects in three ecosystems, A, B and C.

**TABLE 1: RESULTS OF ECOLOGICAL STUDY OF INSECTS
IN THREE ECOSYSTEMS**

Species of Insects	Number of Insects in Each Ecosystem		
	Ecosystem A	Ecosystem B	Ecosystem C
Grasshoppers	50	30	45
Wasps	35	3	44
Pond flies	18	19	23
Beetles	70	63	55
Species diversity	3.39	2.55	

- (i) The species diversity for Ecosystem A and Ecosystem B are given in Table 1. Calculate the species diversity for Ecosystem C using the equation below.

$$D = \frac{N(N-1)}{\sum n(n-1)}$$

[3 marks]

- (ii) Which of the three ecosystems is likely to better withstand ecological disturbance? Give ONE reason for your answer.

[3 marks]

Total 20 marks

2. (a) Outline the role of natural selection in the evolution and adaptation of a species. [3 marks]
- (b) Figure 2 shows the percentage impact of different human activities on an aquatic ecosystem and a terrestrial ecosystem. The impact is measured by the decline in the number of organisms.

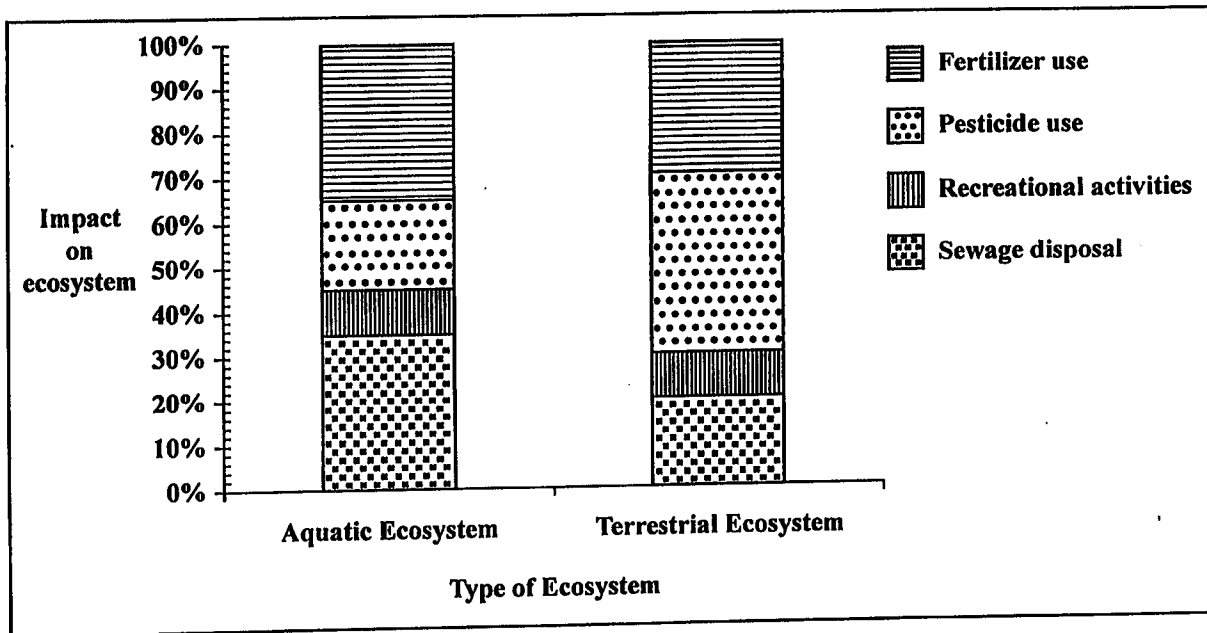


Figure 2. Percentage impact of human activities on an aquatic and a terrestrial ecosystem

- (i) State THREE deductions that can be made from Figure 2. [3 marks]
- (ii) Using values from Figure 2, show which two anthropogenic activities **together** have the **HIGHEST** impact on the terrestrial ecosystem. [2 marks]
- (iii) Explain why the discharge of sewage into aquatic ecosystems should NOT be encouraged. [6 marks]

- (c) Figure 3 shows the relationship between the change in the number of fish and the change in pH in an aquatic ecosystem.

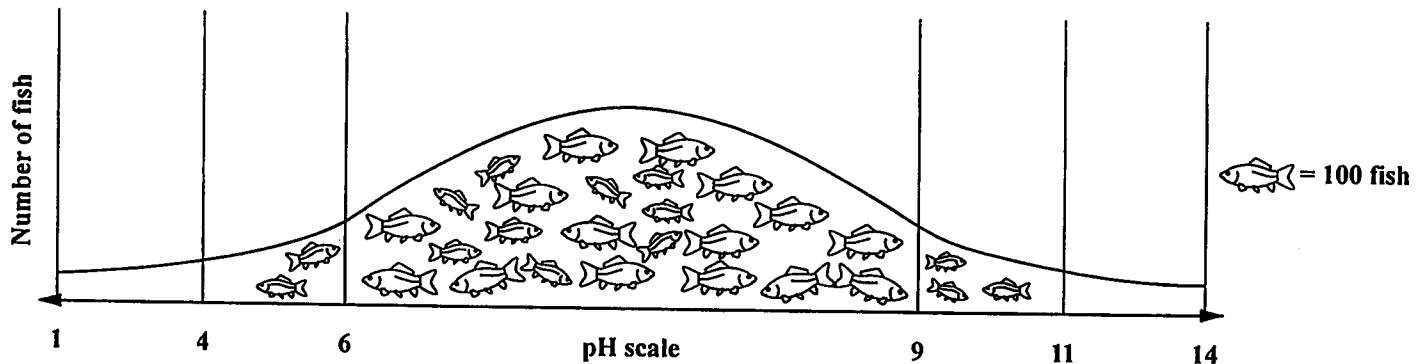


Figure 3. Relationship between the change in the number of fish and the change in pH

Give an explanation for the distribution pattern of fish shown in Figure 3.

[6 marks]

Total 20 marks

MODULE 2

Answer BOTH questions.

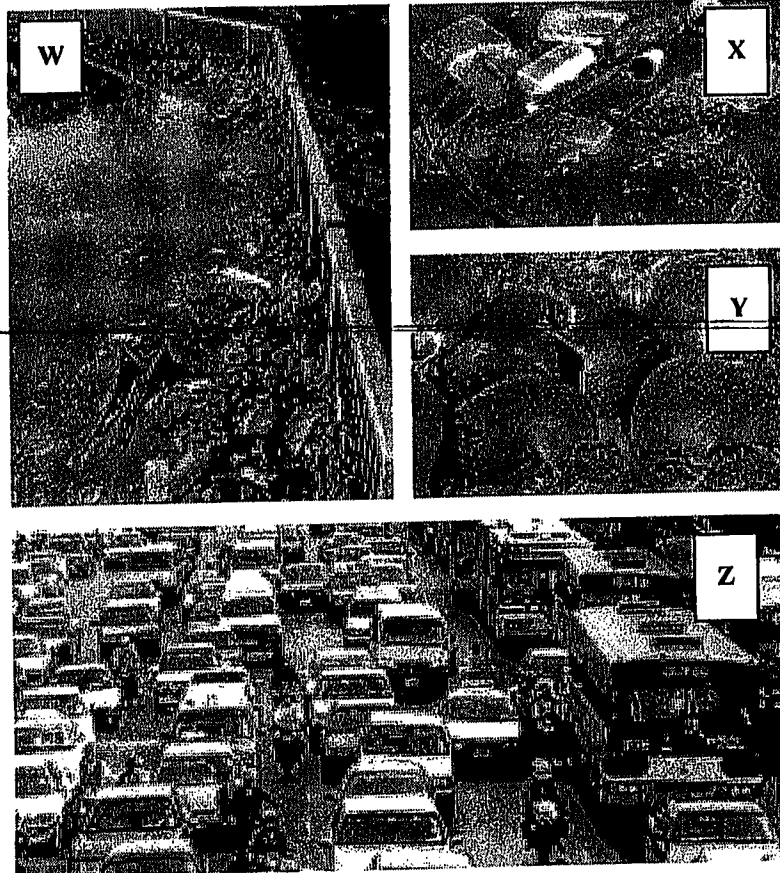
3. (a) Distinguish between Human Development Index (HDI) and Gender Development Index (GDI). [4 marks]
- (b) Explain how EACH of the following factors may affect human population density.
- (i) Climate
 - (ii) Economy
- [6 marks]

- (c) Table 2 shows the percentage of people residing in urban areas in three countries.

TABLE 2: PERCENTAGE OF PEOPLE RESIDING IN URBAN AREAS OF THREE COUNTRIES FROM 1960 – 2000

Country	1960	1980	2000
	%	%	%
A	24	35	55
B	58	65	75
C	72	70	75

- (i) Describe the trend of urbanisation for each country as shown by the data in Table 2. [3 marks]
- (ii) The total population of Country B in 1980 was 7.5 million. Calculate the number of people from Country B who lived in rural areas in 1980. [3 marks]
- (d) The pictures below depict environmental issues associated with urbanization.



- (i) Identify TWO environmental impacts depicted in the pictures. [2 marks]
- (ii) Explain ONE way by which ONE of the impacts identified in (d(i)) may be mitigated. [2 marks]

Total 20 marks

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4. Figure 4 shows the total fertility rates for women in a country, who are educated at different levels.

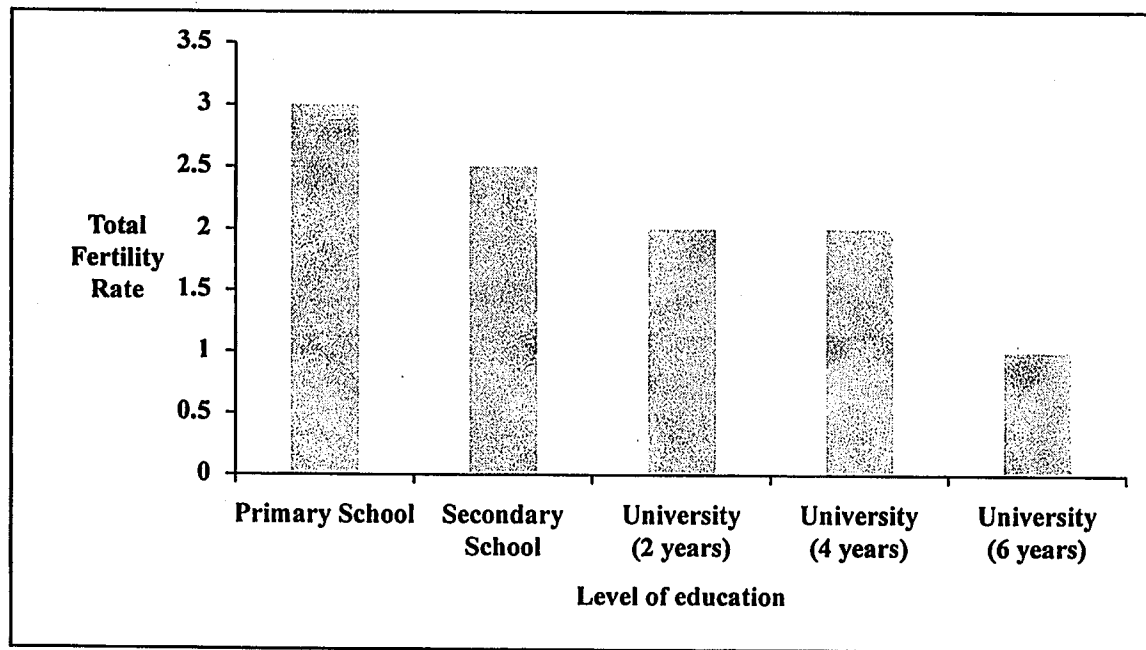


Figure 4. Total fertility rates for women by education levels

- (a) State FOUR deductions that can be made from the graph about the relationship between total fertility rate and the level of education of women. [4 marks]
- (b) Suggest TWO reasons that can explain the relationship between total fertility rate and the level of education of women. [4 marks]
- (c) Explain how an increase in the fertility rate may affect the country's ability to use its natural resources in a sustainable manner. [6 marks]
- (d) Table 3 shows data on the per capita waste production in two countries, A and B.

TABLE 3: PER CAPITA WASTE PRODUCTION

Country	Population Size (millions)	Per Capita Municipal Waste Production (tonnes)
A	8.1	2.1
B	68.5	0.2

- (i) Define 'per capita waste production'. [2 marks]
- (ii) In which country, A or B, does waste production have a greater impact on the environment. Support your answer, giving THREE reasons. [4 marks]

Total 20 marks

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MODULE 3

Answer BOTH questions

5. (a) The policies of a country's government can significantly affect the use of natural resources by citizens. One such policy is the use of tax incentives. In 1978, Country A implemented a policy that allowed a 100% tax refund on the cost of purchasing and installing a solar water heating system. With reference to the data provided in Figure 5, answer the questions that follow.

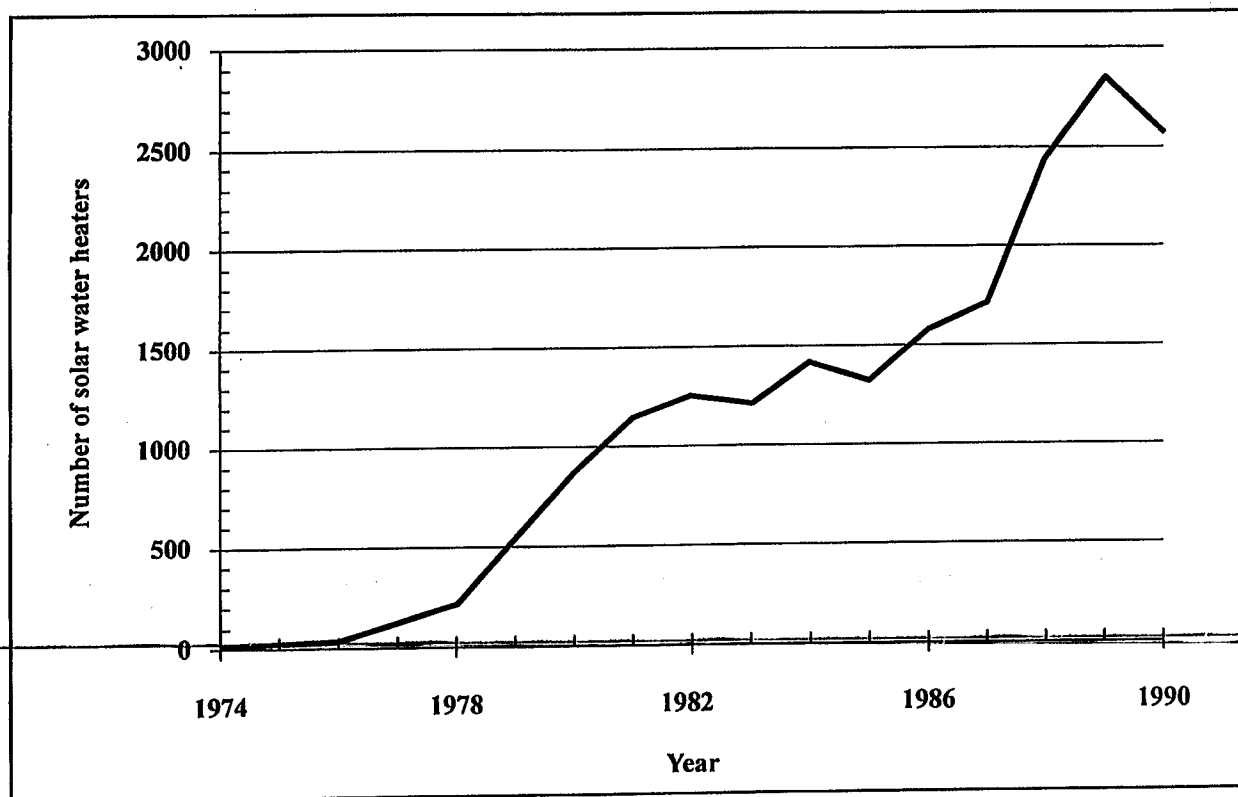


Figure 5: Number of solar water heaters sold in Country A from 1974 to 1990

- (i) Describe how a tax incentive works to manage the use of a natural resource. [3 marks]
- (ii) What was the average annual increase in solar water heaters for the first five years after the tax incentive policy was introduced? [3 marks]
- (iii) Evaluate the effectiveness of the tax incentive for changing the behaviour of Country A's citizens with respect to the use of a natural resource. Use the actual values from Figure 5 to support your response. [6 marks]

- (b) A student wrote an examination in Environmental Science and gave the following INCORRECT answers to a number of questions on the paper. For EACH question, provide the correct answer and explain why the given response is incorrect.

- (i) **Question:** What is the difference between a renewable and an inexhaustible resource?

Answer: There is no difference between these terms, they are the same thing.

[4 marks]

- (ii) **Question:** Explain how technological factors can affect natural resource use in the Caribbean.

Answer: Technological factors do affect natural resource use in the Caribbean. For example, issues such as population growth can have a big impact. As populations increase, they require more raw materials for housing, etc., and so more resources will be used up to satisfy the growing population.

[4 marks]

Total 20 marks

6. (a) A student was overheard making the following statement:

"I don't understand why our teacher says that the landscape is a natural resource. You can't sell a mountain, or make anything out of a beach. I don't think that our teacher is correct about this."

Present TWO arguments in support of the teacher's statement.

[4 marks]

- (b) Using an example, explain what is meant by the 'non-consumptive' use of a natural resource.
[3 marks]
- (c) Give TWO reasons for the importance of ONE named natural resource to a named Caribbean country.
[6 marks]

- (d) Table 4 below provides information on the forest cover of a country.

TABLE 4: LAND AREA UNDER FOREST COVER FROM 1990 TO 2005

Year	Forest Cover (ha)
1990	235 000
1995	233 500
2000	226 000
2005	224 000

- (i) Which five-year period saw the greatest decline in forest cover? Show ALL working. [4 marks]
- (ii) Given that the country has a total area of 512,800 ha, what percentage of the country was under forest cover in 2005? [1 mark]
- (iii) An eco-group in the country is concerned about the rate of loss of forest cover, and they believe that the government should promote natural resource conservation more actively in the country. Give TWO reasons to support the opinion of the eco-group. [2 marks]

Total 20 marks

END OF TEST

IF YOU FINISH BEFORE TIME IS CALLED, CHECK YOUR WORK ON THIS TEST.